Lab 02 – First Contact with OpenPLC

Industrial Security (Bachelor) • Maps to Section 02 • 30–45 min

LAB 02

Goal

Bring up the lab Docker stack and confirm that the soft PLC accepts and runs a ladder-logic program. End the lab with a Modbus read working against it.

Step 1 – Bring up the stack

```
cd bachelor/labs
docker compose up -d
docker compose ps              # all services should be (healthy)
```

Step 2 – Open the OpenPLC web UI

Browse to `http://127.0.0.1:8080`. Log in with the default credentials shown by the OpenPLC project; change the password right away.

Step 3 – Upload the ladder program

Upload the file `scripts/ladder/conveyor.st`. Click **Start PLC**. Open the **Monitoring** page and watch `Motor` toggle when you flip `Start`.

Step 4 – Read holding registers

```
docker compose exec student python3 /labs/scripts/modbus_read.py
```

You should see ten register values. They will all be zero unless the program has written to them.

Deliverable

Hand in: (a) a screenshot of the OpenPLC *Monitoring* page with your running program, and (b) the terminal output of `modbus_read.py`.

Caution

The Modbus listener is bound to `127.0.0.1:5502` on your host. Do not expose it to your network.